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1. Bluetooth Low energy

1.1 General Information

The FT95 is able to transfer stored measurement data via Bluetooth Low Energy. The FT95 always works as GAP peripheral / GATT server.

Unless specified otherwise, all fields are in the order of LSO to MSO. Where LSO = Least Significant Octet and MSO = Most Significant Octet. The device uses 6 Digit PassKey Entry for the connection.

1.2 Advertising Data

The advertising name is "FT95" and the advertising data contains the following information:

Name	Assigned Number	Description	Note
Manufacturer Specific Data	0xFF	0x110601 The first two bytes 0x0611 are the company ID of "Beurer". The 0x01 is used to identify devices that are using the passkey feature	

1.3 Services and Characteristics

Specification Name	Assigned Number	UUID	Note
Generic Access	0x1800	00001800-0000-1000-8000-00805f9b34fb	
Generic Attribute	0x1801	00001801-0000-1000-8000-00805f9b34fb	
Health Thermometer	0x1809	00001809-0000-1000-8000-00805f9b34fb	
Device Information	0x180A	0000180A-0000-1000-8000-00805f9b34fb	
Battery Service	0x180F	0000180F-0000-1000-8000-00805f9b34fb	

Generic Access

Name	Assigned Number	Properties	Note
Device Name	0x2A00	Read	"FT95"
Appearance	0x2A01	Read	"768" – Generic Thermometer
Peripheral Privacy Flag	0x2A02	Read, Write	Privacy is disabled in this device
Reconnection Address	0x2A03	Write	
Peripheral Preferred Connection Parameters	0x2A04	Read	

Generic Attribute

Name	Assigned Number	Properties	Note
Service Changed	0x2A05	Indicate	

Health Thermometer

Name	UUID	Properties	Note
Temperature Measurement	0x2A1C	Indicate	
Measurement Interval	0x2A21	Read	0x00 00

Device Information

Name	Assigned Number	Properties	Note
Serial Number String	0x2A25	Read	SW201805080000
Manufacturer Name String	0x2A29	Read	"Beurer GmbH"
System ID	0x2A23	Read	0000000000000000
Model Number String	0x2A24	Read	FT95
Hardware Revision String	0x2A27	Read	1.0.0
Software Revision String	0x2A28	Read	3.9.2
PNP ID	0x2A50	Read	0x0121

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IEEE	0x2A2A	Read	00
Firmware Revision String	0x2A26	Read	V128

Battery Service

Name	UUID	Properties	Note
Battery Level	0x2A19	Notify, Read	

1.4 Temperature Measurement

The FT95 is sending the data automatically after a connection has been established successfully and the Descriptor of UUID 0x2A1C is set. This can be triggered by clicking the M Button on the device. It is also triggered after every measurement. The device is sending all stored data. If there is a connection between the FT95 and the phone and the Descriptor of UUID 0x2A1C is not set within 30s the FT95 shows an error.

Temperature Measurement – Indicate

Name	Format	Info	Note
flags	1 byte	See separate Flag table "flag description".	
Temperature Measurement	float (4byte)	The 3 lowest bytes are the Mantissa and the high byte is the Exponent e.g. 0xFF00016C means 36.4	
timestamp	time (7byte)	See table Timestamp description – Part of Temperature Measurement for the timestamp data	
Temperature type	1 byte	The temperature type tells the user where the temperature was measured. 0x02 means Body measurement and is used to indicate the forehead measurement, 0xFF means object measurement and is used to indicate an object measurement,	

Example:

Receiving: [0x07 0x10 0x03 0x00 0xFF 0xE1 0x07 0x04 0x0D 0x0E 0x07 0x00 0xFF]

Meaning: 0x07 – The flags set for the FT95 | 0xFF 0x00 0x03 0x10 – Temperature value of 78.4 | 0x07 0xE1 – Year 2017 | 0x04 – Month April | 0x0D – Day 13 | 0x0E – Hour 14 | 0x07 – Minute 07 | 0x00 Second 0 | 0xFF Object Measurement

Flag description – Part of Temperature Measurement

The flags have a size of 1 byte and are part of the temperature measurement Indication.

unit flag	1 bit	0 = Celsius 1 = Fahrenheit	
time stamp flag	1 bit	0 = not present 1 = present	
Temperature type flag	1 bit	0 = not present 1 = present	
reserved for future use	4 bit		
Fever bit	1 bit	0 = no fever 1 = fever	

Timestamp description – Part of Temperature Measurement

The timestamp has a size of 7 bytes and is part of the temperature measurement Indication.

Name	Format	Info	Note
year	uint16 (2 byte)	The value of 1 equals 1 year. The valid range is from 1582-9999	
month	uint8 (1 byte)	The valid range is from 0-12. The value 0 represents unknown, the value of 1 represents January, the value of 2 represents February ...	
day	uint8 (1 byte)	The valid range is from 0-31. The value 0 represents unknown, the value of 1 represents the first day, the value of 2 represents the second day...	
hours	uint8 (1 byte)	The value of 1 equals 1 hour. The valid range is from 0-23	
minutes	uint8 (1 byte)	The value of 1 equals 1 minute. The valid range is from 0-59	
seconds	uint8 (1 byte)	The value of 1 equals 1 second. The valid range is from 0-59	

 health and well-being	Title	Data transmission protocol		
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01	2019-08-02	Initial version		
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